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Analyzing MKR's Underperformance

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MakerDAO is the largest DeFi protocol by value locked (or TVL), with over $2.5 billion in collateral. Maker is also perhaps the oldest DeFi project, pioneering on-chain governance and financial dApps. The supply of DAI, Maker's native stablecoin, crossed the coveted $1 billion threshold recently, and Maker's collateral base grew at an astounding 15% per month between Jan 2018 and Dec 2020.

Yet Maker's native governance token, MKR, underperformed the so-called blue chips by a wide margin during this year's DeFi run up. Let's dive into a few reasons why that might have happened.



Source: TradingView

Competing Against Secondary Markets

First, a short introduction to Maker's design.

Users can deposit any of several accepted tokens in a Maker vault and borrow the DAI stablecoin against the value of those tokens. The protocol charges borrowers a stability fee for borrowing / minting DAI. This fee is variable and subject to the discretion of Maker governance.

The stability fee collected is used to buy MKR from the open market and burn it. This is the primary source of value accrual for MKR.

DAI is a debt credit that's pegged to the value of the US dollar. But if you abstract DAI from the Maker stack and replace it with any other stablecoin, it becomes apparent that Maker's real product is leverage. Most borrowers on Maker deposit assets like ETH and WBTC and use their DAI to yield farm or lever into speculative assets.

Maker has to compete with money market protocols like Aave and Compound to provide loans to leverage-seekers. The problem with that is money markets often give users better terms. Minting DAI on Maker might be cheaper than borrowing DAI, but secondary money markets have a wider range of assets to lend/borrow and marginally higher loan-to-value ratios.

You can mint 1 DAI for every $1.5 of ETH you lock in MakerDAO. On Compound, you only need $1.33 of ETH per DAI. While this seems negligible, the difference is stark for large loans. $100,000 of ETH will allow you to mint 66,000 DAI on Maker but borrow 75,000 DAI from Compound.

On Maker, all loans are disbursed in DAI. But on Aave, you can borrow or lend 6 different stablecoins. This also requires DAI to be liquid on exchanges and close to its peg – a problem the stablecoin continues to face.

The DAI Premium

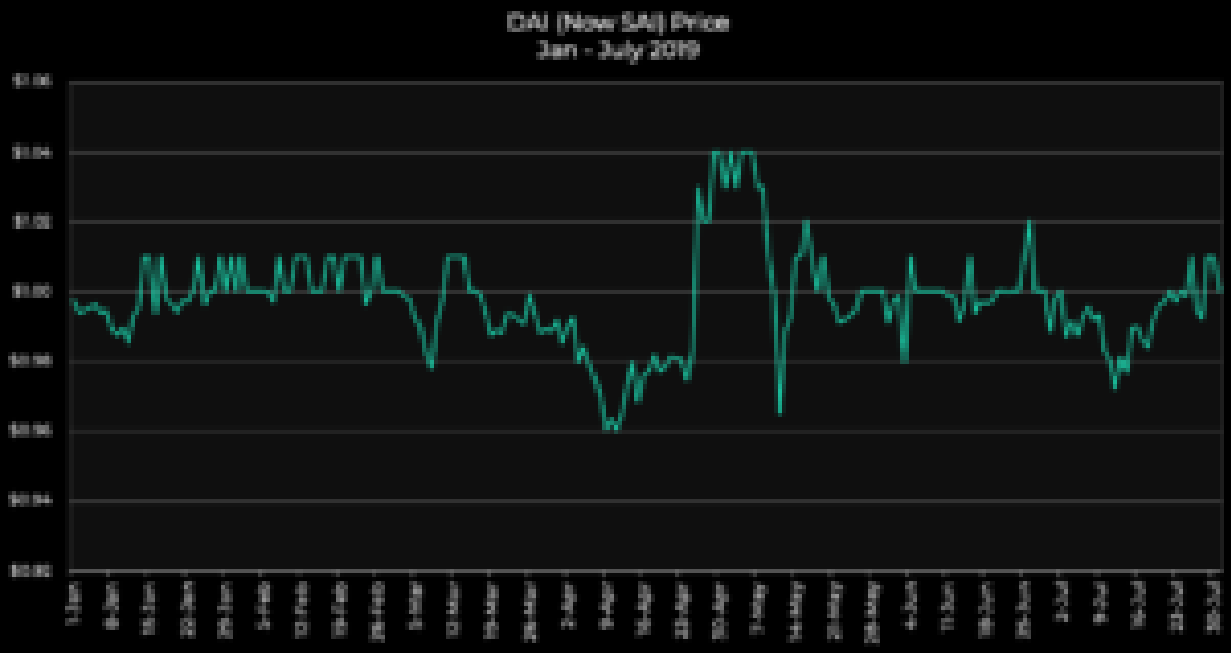
Imagine you just made a profitable trade holding ETH from $400 to $500. You want to take profits and exit with your 25% gain, so you head to Uniswap or wherever else to sell your ETH for stablecoins. You have a few options; you can exit to USDT, USDC, or DAI.

USDT and USDC are trading between $0.995 and $1.005. But you notice DAI is trading at $1.03. If you sell one ETH into USDT or USDC, you receive 497-502 of USDT/USDC per ETH depending on the price. However, selling into DAI nets you only 485 DAI. The obvious choice is to not sell into DAI, which also feeds into why borrowers prefer to use USDT or USDC instead.



Since Multi Collateral DAI, Maker governance has focused its efforts on DAI's peg stability. But Maker's role as a leverage factory impedes this. When DAI is trading below its peg, it needs the market to start buying DAI to prop up the price. Cross-asset arbitrage between DAI and USDC is one part. But peg stabilization also relies on getting existing borrowers to wind down their levered positions, buy DAI below the peg (pushing price up), and close their loan at a slight discount.

In March-April 2019, ETH was exiting a year-long bear market and started to trend upwards. Speculators used Maker to borrow against their ETH holdings, minting DAI and selling that DAI for more ETH. As DAI was minted and sold, it started to de-peg and float below $1. But ETH continued to run and most borrowers didn't want to wind down their positions just to net a few percent in arbitrage profits.



On the other side of the arbitrage is when DAI is trading at a premium, which is the more common scenario. In this situation, the system relies on new capital to borrow DAI and sell it in the open market. Once price returns to $1, the borrower can repurchase their DAI and close the loan. If DAI was minted and sold at $1.04 and purchased and redeemed at $1, the arbitrageur closes a profit of 4% excluding gas fees and slippage.

The problem with this is the uncertainty. Since DAI has sometimes traded at a premium for weeks at a time, the time taken to execute the arbitrage is unknown. It could be a few hours or even a few weeks. Per CoinGecko, in 2020 USDC and USDT cumulatively had 0 days where price closed above $1, while DAI closed over 218 days (62% of the year) with price above $1.

Black Friday and Custodial Collateral

On March 12, 2020, the crypto market witnessed a flash crash that sent BTC from $8,000 to $4,500 in 24 hours. ETH and DeFi tokens fell by much higher percentages, putting an unhealthy amount of borrowers (on Maker, Compound, Aave) at risk of liquidation.

Maker, in particular, suffered a large setback from the flash crash. When a Maker vault reaches the liquidation threshold, keeper bots kick into action. Maker sets up an auction to sell the liquidated collateral and bots source liquidity in DAI and purchase the collateral at a discount to the market. On March 12, over 1200

vaults were liquidated. But when the mass liquidations occurred, Ethereum was so congested that keeper bots were unable to submit any bids at these collateral auctions.

Eventually, some people got their transactions through, but they bid $0 on the collateral instead of the fair value. Since these were the only bids, they won the auction and walked away with free assets. A lot of vaults that were up for partial liquidation were fully liquidated. And since the collateral was given away for free, there was no DAI to pay off the debt of liquidated borrowers.

What ensued was a major headache as Maker faced protocol level losses it needed to recoup, and borrowers lost more money than they should have. The MakerDAO minted and sold 20,980 MKR for 5.3 million DAI (280 DAI per MKR) to cover protocol losses. Later in September, governance voted to reject compensation for vault owners who suffered losses in March, leaving a bad taste in many people's mouths.

Maker quickly added USDC as a collateral in the aftermath of what's now known as "Black Thursday." This would allow the protocol to re-collateralize, increase DAI supply, and improve arbitrage incentives.

Since then, Maker has added more custodial assets like WBTC, TUSD, PAX, and USDT. At the time of writing, custodial assets represent about 30% of MakerDAO's collateral base. DAI's primary value proposition is that it's permissionless and decentralized unlike USDC, USDT, etc. But with centralized assets backing a chunk of DAI's value, there's a valid argument that it erodes DAI's appeal.

From a value accrual perspective, using MKR issuance as a backstop for protocol deficits needs to be accounted for alongside the burns. Looking at the data, from the protocol's inception in Dec. 2017 till Mar. 2020, over 14,500 MKR was burned by the protocol. Black Thursday completely reversed that and more. **Because MKR's primary source of value accrual comes from MKR burns, when Maker's net burn turned into net issuance, MKR became a negative value asset.**



Another problem for Maker is the need to sometimes keep stability fees at 0 for a prolonged period. Ultimately, this makes the protocol's cash flow, and thus MKR's value, a lot less certain.

Since the events of March, Maker and MKR are doing better. After reintroducing stability fees in October, MKR burns have finally re-entered positive territory and current fees owed the protocol imply over 2% of MKR supply can be burned over the next year. Is that enough to drive the price higher? Perhaps not.

What's Next for MakerDAO?

Despite being back on track to regain lost cash flows, I think MakerDAO needs to move away from the buy & burn model. The first step towards regaining the trust of users post Black Thursday would be to use stability fees to bootstrap an insurance fund that protects protocol deficits and vaults. A similar idea was proposed by my teammate Medio back in March after the incident occurred.

If users knew there was a built-in insurance component, like in Compound or the latest iteration of Aave, they would feel much more comfortable borrowing from Maker. This would back into MKR's value too, since the risk of being diluted is reduced. Once a sizable pool has been set up, fees can gradually start funding a buy & burn again, or whatever else governance votes for.

The community has recently started to address MKR's underperformance and question the value accrual model. Acceptance is a necessary first step for improving Maker. The buy & burn model may not be the sole reason for MKR's underperformance, but it's certainly a contributing factor. The need to keep stability fees at 0 for extended periods also didn't help.

I still think MakerDAO has a long way to go in terms of improving its overall design, but as the OG of DeFi, and with a strong community backing it, Maker governance can reform the protocol for the better. MKR, DAI and the DeFi sector in general would benefit from this.